Do Birds of a Father Flock Together? Examining Religious Homophily Dynamics at a Catholic University

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Abstract

In a context where the distribution of a salient dimension of association is skewed toward one group, are minorities more likely to band together (minority distinctiveness) or are majorities more likely to do so (majority exclusiveness)? We address this question with longitudinal network data from the *NetHealth* Cohort Study at the University of Notre Dame ($N = 722$ students), a network ecology in which religious affiliation is a salient dimension of association. Using a dyadic logistic regression framework with an opportunity offset and ego-clustered standard errors, we adjust for physical and social foci, such as residential proximity, and for socio-demographic homophily. The answer depends on how the opportunity baseline is defined, and we report two baselines as co-primary. When the baseline is the composition of nominated student alters, non-Catholic students show higher same-religion tie odds than Catholic students (a single non-Catholic indicator: $\beta = 0.63$, SE $= 0.17$), and this holds after adjusting for ego activity and alter popularity. When the baseline is the religious composition of the ego cohort, the same contrast is near zero ($\beta = 0.04$, SE $= 0.17$). The alter-pool baseline is partly determined by the ties being modeled, so neither baseline is the true opportunity structure. We therefore find the Minority Distinctiveness pattern at the Catholic/non-Catholic level only under the alter-pool baseline, and we find no reliable ranking among Protestant, No Religion, and Other Religion students under either baseline.

1 Introduction

1.1 Background

How does the numeric composition of a local social environment shape the formation of friendship ties across religious groups? While the reliable tendency for individuals to associate with similar others—homophily—is one of social science's most well-established empirical generalizations (McPherson et al., 2001), explaining *why* different groups exhibit varying degrees of homophily within a given population remains an open question (McFarland et al., 2014). Although selectively forming ties with similar others can be an individual choice, it is also influenced by structural factors such as propinquity, opportunity structure, and social foci (Rivera et al., 2010; Feld and Grofman, 2009).

In this last regard, network analysts distinguish between "choice homophily" resulting from individual preferences, proclivities, and relational incentives and "induced homophily" resulting from structural processes such as uneven group sizes or physical proximity, which makes partners of a certain kind more available than others (Kossinets and Watts, 2009; McPherson et al., 2001; Marsden, 1988; Hofstra et al., 2017). Structural clustering can also emerge indirectly via "consolidation" or multiform heterogeneity (Blau, 1977a), in which preferences for similarity along one dimension (e.g., gender or dormitory) narrow the pool of available alters, thereby altering apparent homophily along other dimensions.

In relation to the choice/structural homophily distinction, two major contemporary theories make somewhat divergent predictions about the effect of skewed group sizes on the strength of homophily tendencies. One theories hews more closely to the "agency" (choice homophily and relational incentives) side, and the other comes closer to the structural (strength in numbers) side.

Under the *Minority Distinctiveness Hypothesis* (MDH), a numeric minority status increases the cognitive salience of that group identity in the context of outgroup numerical dominance, which is theorized to produce active sorting patterns consistent with minority members banding together to preserve their subcultural boundaries (Mehra et al., 1998; Leonard et al., 2008; Marsden, 1988; Sepulvado et al., 2015).

Conversely, following Blau's classic treatment, what we might refer to as the *Majority Exclusiveness Hypothesis* (MEH) suggests that a dominant majority possesses the structural and social leverage to guard group boundaries, which means that they should exhibit stronger tendencies towards homophily, doubly aided by their standing as more numerous and more readily available tie-formation partners (majority members are more numerically available) and "induced" (there are more members of the majority to choose from) considerations. As Blau observed, "the larger of two groups discriminates more than the smaller against associating with members of the other group," resulting in higher active homophily among the majority (Blau, 1977a, p. 81).

1.2 Study Setting

An ideal setting for examining the MDH and MEH predictions would be a highly specific network ecology in which a single group holds overwhelming numerical dominance. The University of Notre Dame (ND) presents such a unique social laboratory for the present research (Hachen et al., 2022).

In the *NetHealth Study*'s (Purta et al., 2016) longitudinal cohort, Catholic students represent approximately 72.7% of the NetHealth cohort's egos, while Protestant (9.8%), non-religious (12.3%), and other religious students (4.6%) constitute small, distinct minorities. However, this setting introduces a profound institutional focus (Feld, 1981; Feld and Grofman, 2009): religion is not merely a demographic attribute; it defines the university's very identity and physical infrastructure. Crucially, Catholics attending a Catholic university might be more motivated to befriend

other Catholics because the environment "summons" their religious identity, increasing its salience (Tavory, 2016). Furthermore, ND features built-in residential and social foci—such as single-sex dormitories, mandatory dorm assignments, and Catholic-oriented campus activities—that could organically channel Catholics into homophilous friendships without requiring any active individual preference for Catholic friends; our tie-formation data cannot itself establish whether such preferences are also present, only whether observed mixing patterns exceed what these structural mechanisms would predict on their own.

To evaluate the predictions of the MDH (higher homophily rates among minorities) versus the MEH (higher homophily rates among the majority), it is important to move beyond the limitations of raw homophily rates and aggregate indices. Traditional aggregate measures, such as Krackhardt's E-I (External-Internal) index (Krackhardt and Stern, 1988), suffer from a well-known artifact: in a bounded network, a numeric minority is structurally forced to have higher outgroup contact, regardless of their personal preferences, a phenomenon also emphasized by Blau (1977b), thus producing a form of "induced heterophily." Consequently, aggregate comparisons are biased (Everett and Borgatti, 2012). Here, we address this issue by transitioning to a dyadic (tie-level) framework with an opportunity offset and moving toward margin-free measures of group-level homophily, such as Yule's Q (Borgatti et al., 2024). This approach allows us to adjust for baseline group sizes in the data, as well as other important drivers of homophily, such as physical propinquity and social foci, and for other forms of homophily beyond the focal one in our study.

To this end, this paper leverages 8 waves of longitudinal network data from the *NetHealth* Study to examine the trajectory of religious homophily. We test whether the observed homophily of Protestants and other religious minorities is merely a byproduct of residential sorting and alternative tie-level demographic similarities, or whether it reflects a recurrent, active sorting process consistent with the identity-salience mechanism amid Catholic numerical dominance, as proposed by the MDH. The opportunity baseline is the central modeling choice, and we report the two defensible baselines side by side. Our tie-level data can show how far the results depend on that choice, but cannot directly observe salience or preference as psychological states.

2 Methods and Data

2.1 Data Source and Sample Design

We evaluate the dynamics of religious homophily and friendship choice using longitudinal social network and survey data from the *NetHealth* Study at the University of Notre Dame, a multi-year, NIH-funded data-collection project. *NetHealth* tracked a single cohort of undergraduate students from their entry as freshmen in Autumn 2015 through their graduation in Spring 2019. Initial recruitment enlisted around 700 students (yielding a total sample of $N = 722$ participants who took at least one survey), representing roughly 36% of the typical incoming freshman cohort. While this is an exceptionally high coverage rate for an intensive longitudinal ego-network study, the multi-tiered recruitment strategy (which included snowball sampling) may introduce selection bias into homophily estimates if students systematically recruited highly similar alters into the study. The project tracked their social networks, behaviors, and attributes over 8 waves of data collection fielded approximately every 3–4 months.

In each of the eight waves, respondents completed a name-generator survey with the prompt: "We would like to know who you consider to be in your social network. In the spaces below, please list up to 20 people with whom you spend time communicating or interacting." Beginning with Wave 3, respondents were presented with their previously nominated alters and were allowed to add up to 5 more, for a maximum of 25 alters. These name generators, coupled with name interpreters

detailing each alter's characteristics and the nature of the relationship, yielded a unique multiwave, long-term ego-network dataset that constitutes our primary data source. For the purposes of evaluating within-cohort homophily, we restrict alter nominations to those designated as fellow on-campus students.

2.2 Variables

2.2.1 Ego-Level (Participant) Measures

Ego-level attributes are drawn from the *NetHealth* Basic Survey, which captured demographic and behavioral characteristics across the eight waves. Baseline traits were measured at Wave 1. The primary independent variable is baseline *ego religion*. Respondents were categorized into four groups: Catholic (72.7%, the dominant majority), No Religion (12.3%), Protestant (9.8%), and Other Religion (4.6%). Missing responses account for $< 1\%$. Additional demographic controls include *ego gender*, coded as Woman (51.1%) or Man (47.9%),[1] and *ego race/ethnicity*, categorized into White (65.0%), Latino/a (12.3%), Asian-American (8.7%), Foreign Student (7.1%), African-American (5.7%), and Other ($< 1\%$). To control for individual-level *religious salience (frequency of discussion)* independent of homophily, we use a measure of how frequently the respondent discusses religion (captured in Wave 2). Responses are recorded on a 5-point Likert scale ranging from "Not at all" (4.6%), "Less than 1-2 times a month" (11.8%), "1-2 times a month" (31.9%), "1-2 times a week" (27.4%), to "Three times a week or more" (12.6%).

2.2.2 Alter-Level (Tie) Measures

Alter characteristics and relationship contexts are derived from the *NetHealth* Network Survey, in which egos provided information about their nominated peers. For the *alter relationship context*, denoting the relation of the alter to the university (e.g., Student, Alum, Prospect), 48.6% of overall nominations were for fellow students. We subset our dyadic data to these alters ($N = 17{,}452$ student nominations), then drop 312 on-campus family nominations (e.g., siblings who both attend the university), leaving $N = 17{,}140$ student ties across Waves 1–8 (the regression analyses use Waves 3–8). Egos reported the perceived *alter religion* of each nominated fellow student (Catholic: 67.4%; Protestant: 6.9%; No Religion: 7.2%; Other Religion: 2.3%; Missing: 16.2%). This is used to construct our primary outcome variable, a binary indicator of religious homophily (coded 1 if ego and alter share the same religious affiliation, and 0 otherwise). Egos also reported each student *alter gender and race* (e.g., Woman: 52.2%; Man: 47.8%; and White: 77.8%, with 7.5% missing). These allow us to construct binary dyadic indicators for gender homophily and racial/ethnic homophily. To adjust for structurally induced clustering, egos indicated whether each fellow student shared *physical and social foci (roommates and dorms)*, noting whether the alter was a roommate (13.7% yes) and whether they resided in the same dormitory (35.3% yes); each item is missing for 31.2% of ties.[2] Finally, egos were asked to report how close they felt to each nominated alter on an ordered scale of *friendship closeness (intimacy)*. We construct an ordered numeric indicator for friendship intensity (coded 1 for "Distant", 2 for "Less Than Close", 3 for "Merely Close", and 4 for "Especially Close"). This continuous indicator is added to our multivariate models to control for baseline differences in the homophily of strong versus weak ties. In the regression samples, ties with missing roommate, dorm, or alter-race information are removed by complete-case filtering, which removes disproportionately many No Religion egos (Section 3.3.5); a missing-indicator alternative is reported there.

[1] The survey codes are Female and Male; we use Woman and Man in the text.

[2] At Notre Dame, the dormitory system constitutes a primary axis of residential and social life.

2.3 Opportunity Offset Formulation

To model the likelihood that a tie is homophilous on religion while adjusting for baseline group proportions, we employ a dyadic logistic regression with an opportunity offset (Scott and Wild, 1991). In highly skewed settings like Notre Dame, where Catholics make up nearly three-quarters of the NetHealth cohort, a traditional regression approach struggles to separate active friendship preferences from simple numeric availability. A Catholic student is highly likely to befriend another Catholic simply because the vast majority of their available peers are Catholic (Blau, 1977b), not necessarily because they are actively seeking out a same-religion friend. The opportunity offset corrects for this by subtracting the baseline probability of meeting someone of the same religion under conditions of random mixing.

Formally, we specify the model as:

$$\text{logit}(P(\text{Same Religion} = 1)) = \beta_0 + \beta_1 \text{EgoReligion} + \mathbf{X}\beta + \text{opportunity_offset} \tag{1}$$

In this equation, the opportunity_offset is defined as $\log\left(\frac{p_g}{1-p_g}\right)$, where p_g is the baseline share of the respondent's religious group. We use two baselines, reported as co-primary:

- **Ego-cohort baseline.** p_g is the share of group g among all 718 egos with religion known (Wave 1 religion). It is constant across waves and does not depend on the ties being modeled. It reflects the composition of the recruited sample, not of the student body.
- **Alter-pool baseline.** $p_{g,w}$ is the wave-specific share of group g among nominated student alters. This pool is built from the same nominations being modeled, so it is partly determined by the outcome. Catholics are about 80% of the pool but 73% of egos, which is the difference that drives the sensitivity results below.

By forcing this term into the model with a fixed coefficient of 1.0, we adjust the baseline odds of tie formation. Consequently, the remaining coefficients represent the *excess* tendency to form a homophilous tie above and beyond what the structural opportunity pool dictates, under the chosen baseline.

The model also includes $\mathbf{X}$, a vector of dyadic controls representing alternative mechanisms that might induce homophily without active religious sorting. These include demographic similarities (same gender and same race) as well as physical and social foci (being roommates or living in the same dorm). Our main test is whether a non-Catholic group's coefficient is positive and distinguishable from the Catholic coefficient, under each baseline. A positive coefficient for a minority group indicates that the group exhibits stronger religious homophily than the Catholic majority after accounting for group size and alternative sorting mechanisms. Under the MDH, that pattern would indicate that numeric minorities form tighter in-group bonds in response to a dominant-majority environment. Whether the pattern reflects this mechanism or a baseline artifact is what the sensitivity analyses below address.

Appendix A details the methodological advantages of this dyadic approach relative to two natural alternatives: aggregated E-I index measures, and mixed-effects multilevel models.

3 Results

3.1 Preliminary Checks

3.1.1 Opportunity Structure Stability

Before analyzing active homophily, it is crucial to establish the stability of the baseline opportunity structure. If the numeric composition of the student body fluctuated wildly over the four years, any changes in tie formation could simply be an artifact of changing demographics rather than evolving social preferences. The wave-specific proportions of available student peers across Waves 3 to 8 are plotted in Figure 1. The wave-specific shares are nearly constant: Catholic students make up 77–82% of the alter pool in each wave, No Religion 7–11%, Protestant 8–10%, and Other Religion 2–3%. This stability makes it unlikely that changes in the homophily estimates in subsequent models reflect demographic drift rather than behavioral sorting. The alter-pool shares are stable across waves but differ from the ego-cohort shares, which is why the two baselines give different results.

3.1.2 Baseline Mixing Patterns (Yule's Q)

To address the group-size artifact, we first evaluate the longitudinal trajectory of religious mixing using group-level, baseline-adjusted Yule's Q across all eight waves of the college cohort (see Figure 2). Unlike alternative aggregate metrics—such as raw in-group tie percentages, Coleman's homophily index, or Krackhardt and Stern's E-I (External-Internal) index—Yule's Q is independent of relative group sizes (Borgatti et al., 2024, 159). Because traditional measures like the E-I index are strictly bounded by the available opportunity pool, they inherently force numeric minorities to exhibit higher rates of heterophily simply because of their small baseline population, thereby distorting cross-group comparisons (for a review mapping segregation measures to theoretical properties, see Bojanowski and Corten, 2014).

In contrast, Yule's Q is a direct transformation of the odds ratio, so it is margin-free: its expected value under random mixing is 0 for each group, whatever the group's size. The observed Q is computed from the tie table alone and does not depend on the opportunity baseline. The null distribution against which it is tested does depend on the baseline (see Section 3.3). Margin-invariance of the expected value does not imply that raw Q magnitudes are comparable *across* groups of very different sizes. A degree-preserving permutation analysis shows that, for groups as small as Other Religion, the null distribution of Q is itself wide and can be biased away from zero, so a given raw Q is far less surprising for a small group than the identical value would be for a large one. We therefore use Yule's Q to establish that each group, individually, sorts more than its own chance baseline predicts, rather than to rank the relative strength of homophily across groups of unequal size.

Yule's Q is defined as:

$$Q = \frac{ad - bc}{ad + bc} \tag{2}$$

Where a is the number of ingroup ties (e.g., Catholics befriending Catholics), b is the number of outgroup ties from group G (Catholics befriending non-Catholics), c is the number of ties from the outgroup to group G (non-Catholics befriending Catholics), and d is the number of ties strictly between outgroup members (non-Catholics befriending non-Catholics).

The resulting index ranges from -1 to $+1$. A value of 0 indicates perfect random mixing, in which ties form without regard to religion. A positive value ($Q > 0$) indicates "inbreeding

homophily" (Marsden, 1988): an excess of same-group ties relative to what random mixing would predict, consistent with active sorting toward same-group friends. At the extreme, a Q of $+1$ implies absolute homophily, where group members only ever befriend their own kind. Conversely, a negative value ($Q < 0$) indicates heterophily: an excess of cross-group ties relative to chance, consistent with active avoidance of same-group ties, with -1 indicating that every friendship crosses religious lines.

Yule's Q is positive for every religious group in every wave. Descriptively, this is consistent with *universal inbreeding homophily* ($Q > 0$): over the four years of college, each group's same-religion friendships exceed the rate expected under the chance baseline. The permutation test below indicates that this departure is statistically significant in all but one group-wave under both nulls. Because the null is baseline-dependent, the claim holds only conditionally on each baseline.

Rather than relying on the raw magnitude of Q alone, we anchor this claim to a formal permutation test against each group's chance baseline, presented alongside our other cross-group precision diagnostics in Section 3.3.

3.2 Primary Analysis: Pooled Longitudinal Regressions with Robust Clustered Standard Errors

Our primary test pools ties across the six waves for which the full covariate battery is available (Waves 3–8), rather than resting on any single cross-section. We fit a nested series of pooled models and an interaction model under each of the two opportunity baselines (Table 1, ego-cohort baseline; Table 2, alter-pool baseline). Because individual students are observed repeatedly, we estimate robust standard errors clustered at the respondent level. The analytic sample is the complete-case sample of 8,562 ties from 422 egos.

Under the alter-pool baseline, all three non-Catholic groups have a positive coefficient that is significant in each of Models 1–3: No Religion ($\beta = 0.553$, $p = 0.012$ in Model 1; $\beta = 0.553$, $p = 0.010$ in Model 3), Protestant ($\beta = 0.479$, $p = 0.016$ in Model 1), and Other Religion ($\beta = 1.694$, $p = 0.001$ in Model 1). The Catholic intercept is $\beta = 0.281$ ($p < 0.001$). Under the ego-cohort baseline, none of the non-Catholic coefficients differs detectably from the Catholic reference: No Religion $\beta = -0.170$ ($p = 0.44$), Other Religion $\beta = 0.769$ ($p = 0.14$), and Protestant $\beta = 0.006$ ($p = 0.98$) in Model 1, with the same pattern in Models 2 and 3. The Catholic intercept is $\beta = 0.638$ ($p < 0.001$). The two baselines therefore differ in whether the non-Catholic groups stand out from Catholics at all.

The single non-Catholic indicator, which is the cleanest test of the Catholic/non-Catholic contrast, gives $\beta = 0.633$ (SE $= 0.169$, $p < 0.001$) under the alter-pool baseline and $\beta = 0.035$ (SE $= 0.165$, $p = 0.83$) under the ego-cohort baseline. The 95% confidence intervals are about $(0.30, 0.96)$ and $(-0.29, 0.36)$, respectively. Under the alter-pool baseline, this estimate is unchanged by adding ego activity and alter popularity (Model 3, $\beta = 0.632$).

Furthermore, same-dorm co-residence has a negative association with same-religion tie formation under both baselines (alter pool: $\beta = -0.253$, $p = 0.058$, Model 2; cohort: $\beta = -0.249$, $p = 0.063$, Model 2). Race homophily is positive under both (alter pool: $\beta = 0.268$, $p = 0.037$, Model 2; cohort: $\beta = 0.280$, $p = 0.030$, Model 2). In Model 3, the same-dorm and race coefficients remain near these values (dorm $\beta \approx -0.24$, $p \approx 0.06$–0.07; race $\beta \approx 0.26$–0.27, $p \approx 0.03$–0.04).[3]

[3] Adding each ego's log out-degree and the log of each alter's in-degree excluding the focal nomination (Model 3) changes the group coefficients by 0.012 or less under either baseline. Ego activity is positively associated with the outcome ($\beta = 0.127$, $p = 0.19$ alter pool; $\beta = 0.130$, $p = 0.18$ cohort). Alter popularity is not ($\beta = -0.125$, $p = 0.14$ alter pool; $\beta = -0.118$, $p = 0.17$ cohort). Alter popularity excludes the focal nomination, so the degree control does not mechanically contain the outcome tie.

In the interaction models, the group-by-wave terms behave differently under the two baselines. Under the ego-cohort baseline, the No Religion-by-wave term is $\beta = 0.233$ ($p = 0.006$), which implies a contrast of about -0.72 at Wave 3 (the reference wave; $p = 0.007$) that rises to about $+0.44$ by Wave 8. The joint test of all three group-by-wave terms gives $p = 0.056$. Under the alter-pool baseline, the No Religion-by-wave term is $\beta = 0.080$ ($p = 0.35$), and the joint test gives $p = 0.69$. Both interaction models rest on few unique egos in the non-Catholic groups (37 No Religion and 19 Other Religion), so the cohort result should be treated as a hypothesis-generating pattern rather than a precise estimate of change.

To evaluate the group differences on the probability scale, we also compute average marginal effects of each non-Catholic affiliation relative to Catholic. Each wave is a separate cross-sectional logit with the Model 2 covariates, including religious salience. The effect is averaged over all observed ties in each wave, with the opportunity offset fixed at 0. This is a counterfactual in which every group has a 50% share. Figure 3 shows the ego-cohort version and Figure 4 the alter-pool version. Under the alter-pool baseline, the Other Religion estimate is positive and its interval excludes zero in every wave (roughly 0.27–0.34). The No Religion estimate is positive, with intervals excluding zero, in Waves 4, 5, and 8 (roughly 0.13–0.19), and its intervals include zero in Waves 3, 6, and 7. The Protestant estimate has intervals excluding zero in Wave 4 and, borderline, Wave 5 (roughly 0.09–0.16), and includes zero in the other waves. Under the ego-cohort baseline, no group's interval excludes zero except No Religion in Waves 3 and 4, where the estimate is negative (about -0.27 and -0.14). The ordering among the three non-Catholic groups is not a reliable ranking under either baseline.

3.3 Robustness Checks

The descriptive Yule's Q trajectories and the pooled regression coefficients above share a common vulnerability: a statistic can be correctly bias-adjusted for group size and still be far noisier for a small group than for a large one. Bias correction and precision are separate properties, and only the former is addressed by the opportunity offset or by Yule's Q's margin-invariance. The checks below address baseline dependence, precision, and sample composition.

3.3.1 Yule's Q Permutation Test

We anchor the universal-inbreeding-homophily claim to a degree-preserving permutation test: for each ego, we hold degree and own religion fixed and redraw nominated alters' religion from either the ego-cohort composition (cohort null) or the wave's nominated-alter composition (pool null). As Table 3 shows, every group's observed Q lies above its null mean in every wave under both nulls. Under the cohort null the difference is significant at $p < 0.05$ (two-sided) in 31 of 32 group-waves, with Protestant in Wave 1 the exception ($p = 0.126$). Under the pool null, it is significant in 31 of 32, with the same exception ($p = 0.172$). These tests show whether each group departs from its own chance baseline, not whether one group sorts more than another, since a group's null variance scales with its size.

3.3.2 Bootstrap and Wald Tests of Pooled Coefficients

We resampled the unique egos within each group with replacement (2,000 replicates; Table 4, ego-cohort baseline; Table 5, alter-pool baseline). Under the alter-pool baseline, Protestant and No Religion egos show an excess in-group tie rate above their chance baseline, with 95% percentile intervals excluding zero (about 9 and 14 percentage points). The Catholic excess is about 4 percentage points, with an interval excluding zero. Other Religion's excess is about 13 percentage

points, with interval $(0.027, 0.275)$. Under the ego-cohort baseline, the excess is about 8–11 percentage points in every group, and all intervals overlap. The excess rate depends on each group's baseline share by construction, so these differences describe the baseline as much as the groups. We therefore use the log-odds contrasts for group comparisons.

The Wald tests on the pooled log-odds coefficients (Holm-corrected across six pairwise contrasts; ego-clustered vcov) give the following results. Under the alter-pool baseline (Model 1), each Catholic-versus-non-Catholic contrast has an uncorrected $p \leq 0.016$, and Holm-adjusted values are 0.061 (No Religion), 0.006 (Other Religion), and 0.064 (Protestant). The No Religion versus Other Religion contrast has raw $p = 0.040$ (Holm 0.079), Other Religion versus Protestant has raw $p = 0.026$ (Holm 0.078), and No Religion versus Protestant is clearly null ($p = 0.79$). Model 3 gives nearly the same pattern. Under the ego-cohort baseline, no contrast is significant before or after correction (smallest raw $p = 0.09$). The Holm-adjusted Catholic-versus-non-Catholic values under the alter pool are larger than in the earlier complete-case sample because the standard errors are larger in the present sample, while the point estimates are almost unchanged (No Religion 0.553 vs. 0.549 before the sample change).

3.3.3 Baseline Sensitivity

Table 6 reports three diagnostics. First, the single non-Catholic indicator is reported under the ego-cohort baseline, the alter pool, and a leave-own-ego-out pool that removes each ego's own nominations from the pool it is compared against. The leave-own-ego-out pool gives $\beta = 0.638$ (SE = 0.170), essentially the same as the alter pool. The ego's own nominations therefore do not drive the alter-pool result. The remaining source of endogeneity is group-level, because all egos' nominations still shape the pool. Second, the baseline share at which each group's contrast reaches zero is about 10.5% for No Religion (ego-cohort share 12.4%; alter-pool share 9.0%), 9.3% for Other Religion (cohort 4.6%; alter pool 2.7%), and 9.9% for Protestant (cohort 9.9%; alter pool 8.9%). These values are approximate, because they use group-mean offsets with the Catholic share held at its cohort value. The No Religion contrast changes sign between the two baselines, and the Protestant break-even share equals its cohort share, which is consistent with the cohort-baseline null for Protestant. Third, a free offset coefficient on the alter-pool baseline is 0.88 (SE = 0.28), which does not differ from 1 ($p = 0.67$). This supports the offset specification under the alter pool, but the coefficient is identified only from wave-to-wave variation in the pool, which is small. Under the ego-cohort baseline it is not identified, because the offset is constant within group.

3.3.4 Inference With Few Clusters

The Other Religion group has 19 egos in the analytic sample, so cluster-robust standard errors may be unreliable for that group. Table 7 reports two-way standard errors clustered on egoid and alterid (3,011 unique alters), and percentile intervals from an ego-cluster pairs bootstrap (1,000 replicates). Two-way clustering changes the Model 1 p-values only modestly. Under the alter-pool baseline, the No Religion, Other Religion, and Protestant p-values go from 0.012, 0.001, and 0.016 to 0.017, 0.001, and 0.030. Under the ego-cohort baseline, the two-way p-values are 0.47, 0.14, and 0.98. The bootstrap intervals are wide. Under the alter-pool baseline they are $(0.09, 0.98)$ for No Religion, $(0.38, 2.56)$ for Other Religion, and $(0.09, 0.85)$ for Protestant. Under the ego-cohort baseline they include zero for all three groups. Dropping one ego at a time moves the Other Religion alter-pool coefficient between 1.00 and 1.80. The lower value comes from removing a single Other Religion ego who contributes 35 of the group's 292 ties. Under the ego-cohort baseline the same ego moves the coefficient from 0.77 to 0.07. The Other Religion estimate is therefore driven substantially by

individual egos under both baselines. The No Religion and Protestant coefficients are more stable under leave-one-ego-out deletion.

CR2 small-sample standard errors and a generalized estimating equations population-average model are not reported, because the required packages are not yet installed.

3.3.5 Missingness and Sample Composition

The complete-case analytic sample has 8,562 ties, down from 10,046 in the baseline sample. The reduction is not spread evenly across groups. Among No Religion egos, 24.7% report a race outside the four coded categories (Catholic 5.3%, Other Religion 9.1%, Protestant 7.0%). Because same-race homophily is undefined for these egos, the complete-case filter removes them. As a result, the No Religion share of ties falls from 8.5% in the baseline sample to 6.7% in the analytic sample. Table 8 reports Model 1 on the baseline sample and sensitivity models that keep these egos with missing indicators. Model 1 on the full baseline sample gives No Religion $\beta = 0.844$ and Other Religion $\beta = 1.66$ under the alter pool, and 0.104 and 0.716 under the ego-cohort baseline. The missing-indicator models (9,961 ties) give No Religion $\beta \approx 0.80$, Other Religion $\beta \approx 1.64$, and Protestant $\beta \approx 0.56$ under the alter pool, and No Religion $\beta \approx 0.06$ and Protestant $\beta \approx 0.07$ under the ego-cohort baseline, both not significant. The alter-pool magnitudes for No Religion range from 0.55 to 0.84 across these samples and codings, and the cohort-baseline null persists in every version. Ego-level discussion-frequency missingness is 10.7% for Catholic egos, 14.6% for No Religion, 18.2% for Other Religion, and 11.3% for Protestant egos. Alter-religion missingness across nominations ranges from 9.4% (Protestant egos) to 21.8% (Other Religion egos).

3.3.6 Synthesis

Across these checks, the Catholic/non-Catholic contrast is reliable only under the alter-pool baseline. It persists after degree controls, under leave-own-ego-out pool construction, under two-way clustering, and under the missing-indicator coding. Under the ego-cohort baseline, the contrast is not distinguishable from zero in any specification we ran. Neither baseline supports a reliable ranking among Protestant, No Religion, and Other Religion students. The evidence for a Catholic/non-Catholic difference is therefore conditional on the choice of baseline.

4 Discussion and Concluding Remarks

The results do not establish MDH over MEH on their own. Under the nominated-alter baseline, non-Catholic students show higher same-religion tie odds than Catholic students, which is the pattern the MDH predicts. Under the ego-cohort baseline, the same contrast is near zero. Because the alter pool is partly determined by the ties being modeled, the alter-pool result cannot be read as an unbiased estimate of excess homophily, and the ego-cohort baseline does not capture the opportunity structure of the student body either. The Catholic/non-Catholic difference is therefore consistent with MDH only conditional on the alter-pool baseline, and the data do not reject a null effect under the ego-cohort baseline.

The Catholic majority's excess is small under the alter pool (a bootstrap mean of about 4 percentage points, Table 5, Panel A), though reliably above zero. Under the ego-cohort baseline it is about 10 percentage points, and the non-Catholic groups show similar excess rates. The Catholic excess is therefore not interpretable without a choice of baseline, and a description of the Catholic baseline as structural depends on that choice.

Under the alter-pool baseline, the No Religion coefficient is positive and significant in Models 1–3 ($\beta \approx 0.55$, $p \leq 0.012$), while in the interaction model its main effect is not significant ($\beta = 0.353$, $p = 0.20$). Other Religion has the largest coefficient in every specification, but it rests on 19 egos, and dropping one of them reduces the coefficient from about 1.7 to 1.0. Under the ego-cohort baseline, no non-Catholic group is distinguishable from Catholics, and the No Religion coefficient rises across waves from negative to positive. We cannot reliably rank the minority groups under either baseline, and we therefore do not interpret their relative magnitudes.

On change over time, the two baselines disagree. Under the alter pool, group-by-wave terms are not significant (joint $p = 0.69$). Under the ego-cohort baseline, the No Religion-by-wave term is significant ($p = 0.006$, joint $p = 0.056$). Because the ego-cohort offset is constant within group, this trend reflects the outcome alone, and it rests on 37 No Religion egos. We therefore describe the alter-pool result as no detectable change, and the ego-cohort result as a pattern that needs replication with a larger sample. Neither result supports a claim of stable trajectories.

Several limitations remain. First, there is no measured opportunity structure. The student-body composition is not in the data, and the ego-cohort baseline reflects a recruited sample of about 36% of the entering class. Without an external enrollment benchmark, the baseline choice cannot be resolved with the present data. Second, the non-Catholic groups are small. Other Religion has 19 egos in the analytic sample, and single egos can move its coefficient substantially. Third, non-Catholic students have chosen to attend a Catholic university. As a result, they may be less culturally distinct from the Catholic majority than they would be in a secular setting, which may attenuate the boundaries we observe. Fourth, alter attributes are ego-reported, and 16% of nominations lack alter religion. Respondents who do not know a friend's religion may assume that the friend is Catholic, which could inflate Catholic homophily. Fifth, complete-case filtering removes a disproportionate share of No Religion egos. Sixth, the outcome treats directed nominations as separate observations, and reciprocity and the undirected structure of friendship are not modeled. Whether observed mixing reflects constraints on who can reciprocate is not addressed by the present design. Seventh, we observe a single institutional context with one specific distribution of religious groups. As previous research has shown (Goodreau et al., 2009; McFarland et al., 2014), the strength of homophily can vary contextually, often exhibiting curvilinear relationships with group proportions rather than strictly linear responses to minority or majority status. Future work extending these findings across multiple institutional contexts could yield greater theoretical specificity regarding the separable effects of group size and group proportion.

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A Methodological Advantages of the Dyadic Approach

A.1 Benefits of the Opportunity Offset

The Krackhardt and Stern (1988) External-Internal (E-I) index is defined as $EI = (E-I)/(E+I)$. Under random mixing, the expected E-I index for group G is a direct linear function of group size p:

$$EI_{\text{expected}} = (1-p) - p = 1 - 2p \tag{3}$$

Because the baseline shifts with p, raw E-I indices cannot be compared across groups of different sizes to infer preferences. Individual-level aggregations of normalized E-I indices also introduce a severe boundary bias: any minority ego with zero same-religion ties is assigned an EI_{norm} of exactly +1.0. Averaging these individual scores heavily pulls the minority group's average toward +1.0, falsely indicating heterophily. Our dyadic logistic regression with an opportunity offset ($\log(p/(1-p))$) mitigates these biases. Entering this offset with its coefficient constrained to exactly 1.0 subtracts the baseline opportunity structure. Consequently, the model intercept and group coefficients represent the *excess log-odds of homophilous tie formation beyond what is expected under the chosen baseline*. The offset addresses bias in the point estimate due to group size, provided the baseline is correct. It does not guarantee that coefficients are equally *precise* across groups of very different sizes, and the Results show that the choice of baseline can change the estimates substantially. Groups with few unique egos (e.g., Other Religion) yield coefficients with wide uncertainty.

A.2 Why Dyadic Clustered Regression is Preferable to Mixed-Effects Multilevel Models

In social network analysis, egocentric network datasets are inherently hierarchical: nominated friendship ties (Level-1) are nested within individual egos (Level-2 clusters). To correct for this nesting, methodologists frequently recommend mixed-effects logistic regressions:

$$\log\left(\frac{P(Y_{ij}=1 \mid \gamma_i)}{1-P(Y_{ij}=1 \mid \gamma_i)}\right) = \beta_0^C + \beta_1^C \text{EgoReligion}_i + \beta_2^C \mathbf{X}_{ij} + \gamma_i + \text{offset}_{wi} \tag{4}$$

Where $\gamma_i \sim N(0, \sigma^2)$ is a random intercept for ego i, suggesting that multilevel models can be a useful way to model this type of data (Snijders and Baerveldt, 2003). The mixed-effects multilevel model differs conceptually from a population-average dyadic logistic regression with robust clustered standard errors. First, in a mixed-effects logistic model, the estimated fixed effects (β^C) are conditional: they represent the change in log-odds for a *specific individual* if that individual's religion were to change. If religion is a relatively stable baseline trait, as it is in our data, this is a less realistic modeling assumption. In contrast, our model with robust clustered standard errors represents a marginal (population-average) model:

$$\log\left(\frac{P(Y_{ij}=1)}{1-P(Y_{ij}=1)}\right) = \beta_0^M + \beta_1^M \text{EgoReligion}_i + \beta_2^M \mathbf{X}_{ij} + \text{offset}_{wi} \tag{5}$$

Because our core hypothesis concerns group-level differences across the population, the marginal population-average model is the more appropriate target. The conditional and marginal coefficients differ in scale (for a logit model with random intercepts, conditional coefficients are larger in absolute value), so the two are not directly interchangeable. A population-average alternative,

such as generalized estimating equations with an exchangeable working correlation within egos, would be a more direct comparison; it is not reported here.

We do not fit a mixed-effects model to these data, so the comparison above is conceptual. The dyadic model with clustered standard errors does not require a random-effects distributional assumption, but it does not remove the scale difference between conditional and marginal effects noted above.

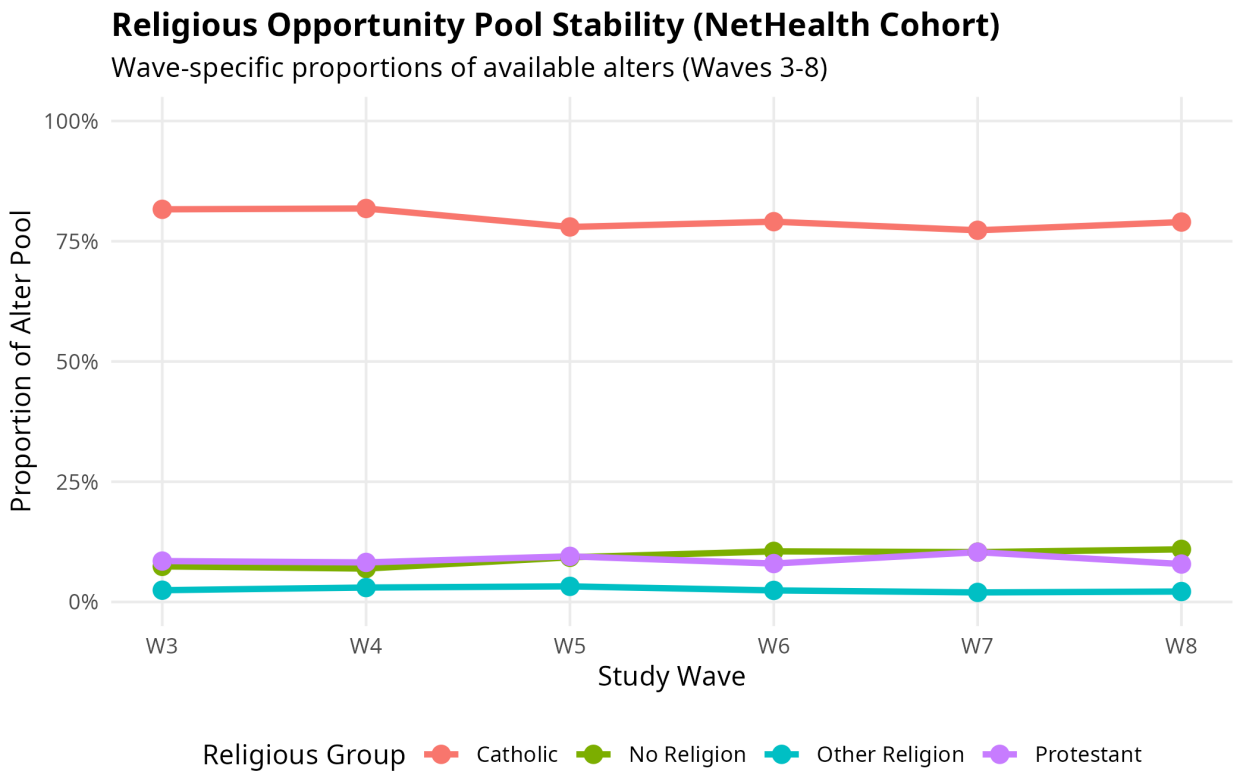


Figure 1: Preliminary Check: Religious Opportunity Structure Stability (Proportions of Available Peers, Waves 3-8).

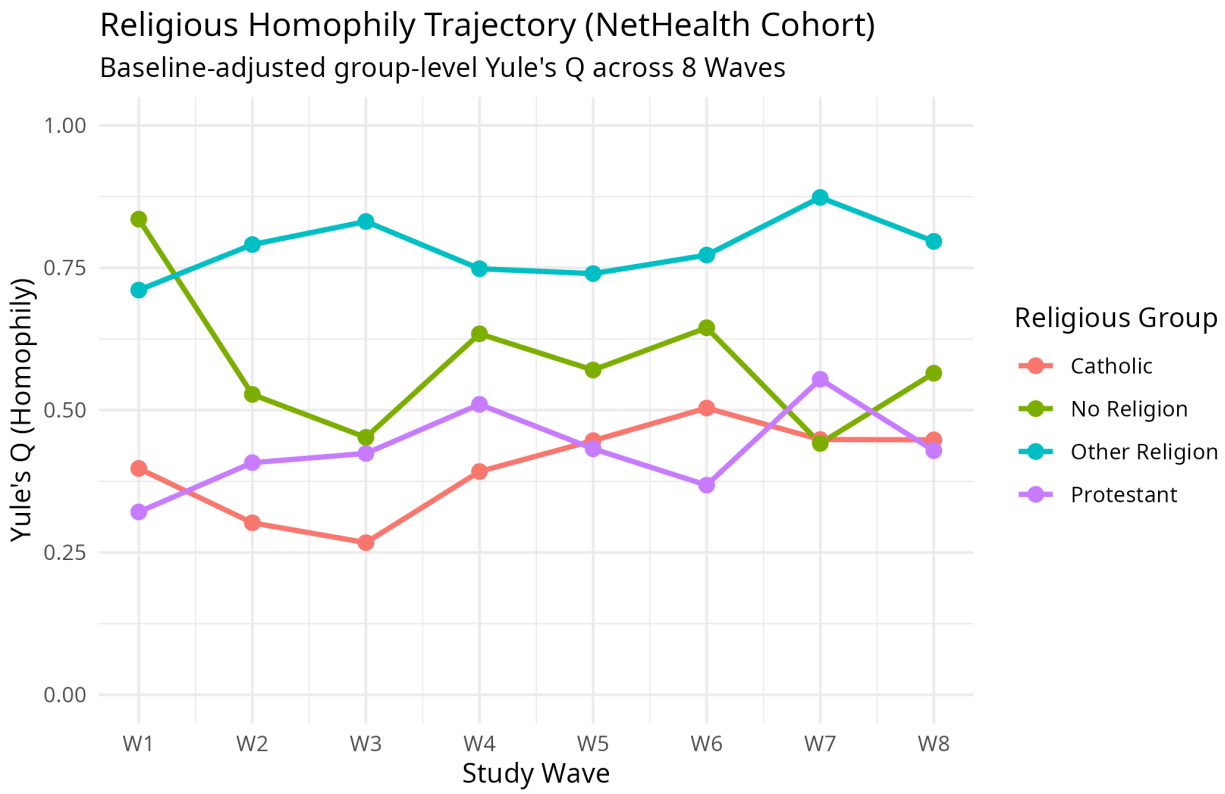


Figure 2: Preliminary Check: Longitudinal Trajectories of Religious Homophily (Yule's Q) across Waves 1 to 8.

Table 1: Primary Analysis: Pooled Longitudinal and Interaction Models, Ego-Cohort Opportunity Offset (Waves 3-8, Clustered SEs).

	Model 1	Model 2	Model 3 (+Degree)	Interaction (+Degree)
Intercept	0.638*** (0.068)	0.569 (0.324)	0.234 (0.432)	0.371 (0.453)
Ego: No Religion	-0.170 (0.222)	-0.169 (0.217)	-0.166 (0.215)	-0.721** (0.269)
Ego: Other Religion	0.769 (0.515)	0.770 (0.508)	0.757 (0.510)	0.698 (0.514)
Ego: Protestant	0.006 (0.200)	0.003 (0.201)	0.007 (0.202)	-0.077 (0.245)
Tie: Gender Homophily	—	0.204 (0.144)	0.196 (0.143)	0.192 (0.142)
Tie: Race Homophily	—	0.280* (0.129)	0.274* (0.129)	0.266* (0.130)
Tie: Roommates	—	-0.160 (0.112)	-0.134 (0.114)	-0.139 (0.114)
Tie: Same Dorm	—	-0.249 (0.134)	-0.243 (0.134)	-0.242 (0.133)
Tie: Subjective Closeness	—	-0.073 (0.078)	-0.061 (0.078)	-0.060 (0.078)
Ego: Religious Salience	—	0.036 (0.061)	0.038 (0.060)	0.032 (0.061)
Ego: Activity (log out-deg.)	—	—	0.130 (0.098)	0.127 (0.098)
Alter: Popularity (log, excl. focal tie)	—	—	-0.118 (0.085)	-0.122 (0.087)
Tie: Wave Number	—	—	—	-0.042 (0.028)
Ego: No Religion x Wave	—	—	—	0.233** (0.085)
Ego: Other Religion x Wave	—	—	—	0.026 (0.119)
Ego: Protestant x Wave	—	—	—	0.036 (0.071)
Joint test, group × wave (3 df), p				0.056
Group sample sizes (unique egos / dyadic ties), Waves 3–8 pooled, fully-adjusted sample				
Catholic	320 / 6,735			
No Religion	37 / 575			
Other Religion	19 / 292			
Protestant	46 / 960			

Note: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$

Egos are counted once even if they contribute ties across multiple waves; standard errors clustered by egoid throughout. On-campus family ties are excluded from the analytic sample. Model 3 and the Interaction model add each ego's log out-degree and the log of each alter's in-degree excluding the focal nomination, computed from the full wave-specific nomination roster. The ego activity term has $p \approx 0.18$–0.20 across Model 3 and the Interaction model. The opportunity offset is the log-odds of each group's share among egos (Wave 1 religion, time-invariant). The alternative offset, based on the composition of nominated alters in each wave, is reported in Table 2.

Table 2: Sensitivity Analysis: Pooled Longitudinal and Interaction Models, Alter-Pool Opportunity Offset (Waves 3-8, Clustered SEs).

	Model 1	Model 2	Model 3 (+Degree)	Interaction (+Degree)
Intercept	0.281*** (0.068)	0.243 (0.325)	-0.080 (0.433)	-0.099 (0.453)
Ego: No Religion	0.553* (0.221)	0.552* (0.217)	0.553** (0.214)	0.353 (0.274)
Ego: Other Religion	1.694** (0.518)	1.696*** (0.509)	1.684*** (0.509)	1.535** (0.513)
Ego: Protestant	0.479* (0.199)	0.476* (0.200)	0.479* (0.201)	0.561* (0.248)
Tie: Gender Homophily	—	0.202 (0.143)	0.193 (0.142)	0.190 (0.141)
Tie: Race Homophily	—	0.268* (0.129)	0.263* (0.128)	0.267* (0.129)
Tie: Roommates	—	-0.159 (0.113)	-0.134 (0.114)	-0.134 (0.114)
Tie: Same Dorm	—	-0.253 (0.133)	-0.247 (0.133)	-0.241 (0.133)
Tie: Subjective Closeness	—	-0.073 (0.078)	-0.061 (0.078)	-0.062 (0.078)
Ego: Religious Salience	—	0.030 (0.061)	0.033 (0.061)	0.031 (0.061)
Ego: Activity (log out-deg.)	—	—	0.127 (0.098)	0.128 (0.098)
Alter: Popularity (log, excl. focal tie)	—	—	-0.125 (0.085)	-0.118 (0.087)
Tie: Wave Number	—	—	—	0.009 (0.029)
Ego: No Religion x Wave	—	—	—	0.080 (0.086)
Ego: Other Religion x Wave	—	—	—	0.063 (0.126)
Ego: Protestant x Wave	—	—	—	-0.035 (0.074)
Joint test, group × wave (3 df), p				0.689
Group sample sizes (unique egos / dyadic ties), Waves 3–8 pooled, fully-adjusted sample				
Catholic	320 / 6,735			
No Religion	37 / 575			
Other Religion	19 / 292			
Protestant	46 / 960			

Note: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$

Egos are counted once even if they contribute ties across multiple waves; standard errors clustered by egoid throughout. On-campus family ties are excluded from the analytic sample. Model 3 and the Interaction model add each ego's log out-degree and the log of each alter's in-degree excluding the focal nomination, computed from the full wave-specific nomination roster. The ego activity term has $p \approx 0.19$–0.19 across Model 3 and the Interaction model. The opportunity offset is the log-odds of each group's share among nominated student alters in each wave. This offset is endogenous to the outcome, since the nominations it is computed from are the ties being modeled. Models use the same analytic sample as Table 1.

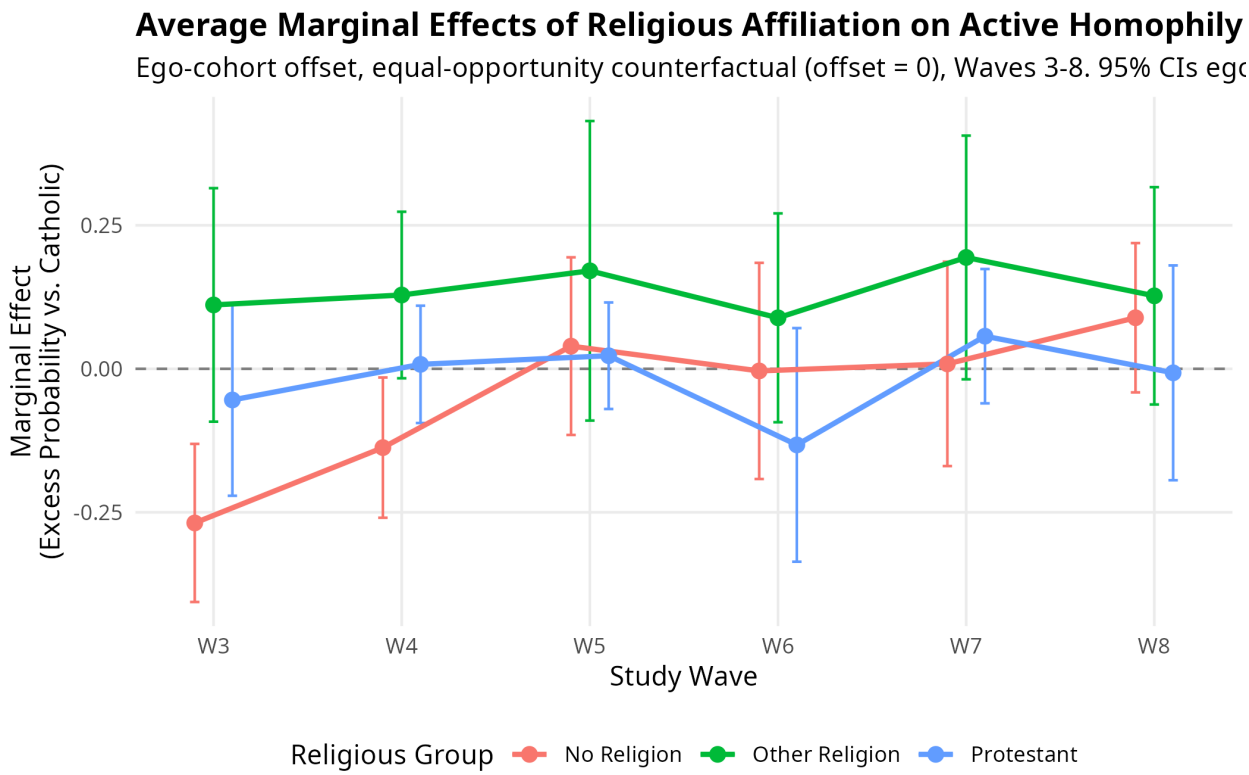


Figure 3: Primary Analysis (ego-cohort baseline): Average Marginal Effects of Religious Affiliation on Active Homophily over Time (Waves 3-8). Each point is the wave-specific average marginal effect, on the probability scale, of belonging to the given non-Catholic group rather than Catholic. Each wave is a separate cross-sectional logistic regression with the Model 2 covariates, including religious salience; the effect is averaged over all observed ties in that wave, with the opportunity offset fixed at 0 (an equal-opportunity counterfactual). Error bars show 95% confidence intervals with ego-clustered standard errors.

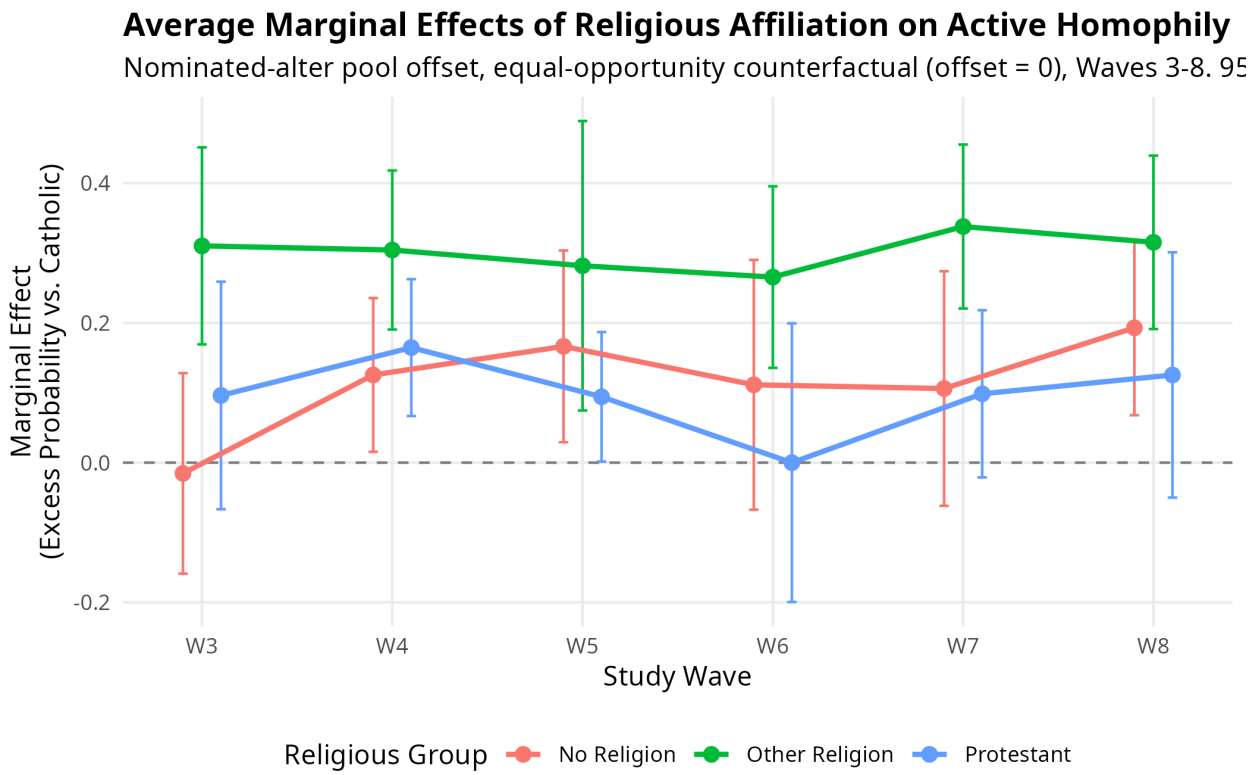


Figure 4: Sensitivity Analysis (alter-pool baseline): Average Marginal Effects of Religious Affiliation on Active Homophily over Time (Waves 3-8). Same specification as Figure 3, with the opportunity offset defined from the wave-specific composition of nominated student alters. Error bars show 95% confidence intervals with ego-clustered standard errors.

Table 3: Robustness Check: Degree-Preserving Permutation Test of Yule's Q (Waves 1–8).

			Cohort null (primary)			Pool null	
Wave	Group	Observed Q	Null Mean	z	p	z	p
W1	Catholic	0.398	-0.006	3.62	<0.001	3.23	0.002
	No Religion	0.835	-0.024	4.27	<0.001	3.39	<0.001
	Other Religion	0.711	-0.282	1.60	0.016	1.62	0.030
	Protestant	0.321	-0.052	1.34	0.126	1.26	0.172
W2	Catholic	0.302	-0.001	7.07	<0.001	6.14	<0.001
	No Religion	0.528	-0.007	6.72	<0.001	5.37	<0.001
	Other Religion	0.791	-0.068	3.00	<0.001	2.45	<0.001
	Protestant	0.408	-0.003	4.58	<0.001	3.96	<0.001
W3	Catholic	0.267	-0.000	4.51	<0.001	3.96	<0.001
	No Religion	0.452	-0.006	3.91	<0.001	3.12	<0.001
	Other Religion	0.831	-0.088	2.54	<0.001	2.00	<0.001
	Protestant	0.424	-0.006	3.57	<0.001	3.37	<0.001
W4	Catholic	0.392	0.001	6.60	<0.001	5.74	<0.001
	No Religion	0.634	-0.013	4.94	<0.001	4.15	<0.001
	Other Religion	0.748	-0.065	2.62	<0.001	2.30	<0.001
	Protestant	0.510	-0.008	4.31	<0.001	4.05	<0.001
W5	Catholic	0.446	0.000	7.59	<0.001	7.24	<0.001
	No Religion	0.570	-0.010	4.64	<0.001	4.05	<0.001
	Other Religion	0.740	-0.061	2.62	<0.001	2.26	<0.001
	Protestant	0.432	-0.008	3.88	<0.001	3.71	<0.001
W6	Catholic	0.504	-0.002	7.61	<0.001	7.46	<0.001
	No Religion	0.645	-0.011	4.89	<0.001	4.66	<0.001
	Other Religion	0.772	-0.090	2.55	<0.001	1.98	<0.001
	Protestant	0.368	-0.008	2.97	<0.001	2.80	0.002
W7	Catholic	0.448	-0.003	6.59	<0.001	6.04	<0.001
	No Religion	0.442	-0.019	2.97	<0.001	2.65	<0.001
	Other Religion	0.874	-0.113	2.71	<0.001	2.01	<0.001
	Protestant	0.554	-0.007	4.42	<0.001	4.32	<0.001
W8	Catholic	0.448	0.000	4.84	<0.001	4.49	<0.001
	No Religion	0.565	-0.016	3.15	<0.001	2.87	0.002
	Other Religion	0.796	-0.169	2.02	<0.001	1.74	<0.001
	Protestant	0.429	-0.027	2.34	0.004	2.05	0.006

Note: Null distributions (1,000 permutations per wave and per null) hold each ego's tie count and own religion fixed and redraw nominated alters' religion from either the cohort composition of egos (primary) or the wave's nominated-alter composition (pool). Observed Q is computed from the tie table alone and is the same under both nulls. Observed Q lies above its cohort-null mean in 32 of 32 group-waves. Two-sided permutation p-values (share of null draws at least as extreme as observed, doubled) are significant at $p < 0.05$ in 31 of 32 group-waves under the cohort null (exceptions: Protestant (Wave 1, $p = 0.126$)) and in 31 of 32 under the pool null (exceptions: Protestant (Wave 1, $p = 0.172$)). Because null variance scales with group size, z-scores should not be compared across groups to infer a ranking of homophily strength.

Table 4: Robustness Check: Ego-Level Cluster Bootstrap and Wald Tests, Ego-Cohort Opportunity Baseline (Waves 3–8, 2,000 replicates).

Panel A: Group-level excess in-group tie rate (descriptive)

Group	n Egos	n Ties	Observed Rate	Baseline Rate	Excess (95% CI)
Catholic	348	7,630	0.835	0.731	0.104 (0.086, 0.120)
No Religion	52	850	0.226	0.124	0.101 (0.045, 0.165)
Other Religion	22	381	0.155	0.046	0.107 (0.007, 0.252)
Protestant	50	1,185	0.181	0.099	0.084 (0.037, 0.134)

Note: Baseline: ego-cohort composition (primary). The excess is a descriptive quantity: for a fixed in-group odds ratio it varies with the group's baseline share, so it should not be compared across groups of different sizes. Panels C–D (log-odds) are the basis for group contrasts. The bootstrap uses the baseline sample (no covariate filtering).

Panel B: Pairwise differences in excess rate (descriptive)

Group 1	Group 2	Mean Diff.	95% CI	Excludes 0
Catholic	No Religion	0.002	(-0.062, 0.062)	No
Catholic	Other Religion	-0.003	(-0.147, 0.099)	No
Catholic	Protestant	0.020	(-0.034, 0.070)	No
No Religion	Other Religion	-0.006	(-0.159, 0.116)	No
No Religion	Protestant	0.017	(-0.058, 0.098)	No
Other Religion	Protestant	0.023	(-0.094, 0.174)	No

Note: Unique egos are resampled with replacement, separately within each group, so that each resampled ego's ties move together. Panel B differences are valid because each group's replicates are drawn independently.

Panel C: Wald tests of pairwise equality, pooled log-odds coefficients (Model 1)

Group 1	Group 2	Log-Odds Diff. (G1 − G2)	SE	χ^2	p	Holm p
Catholic	No Religion	0.170	0.222	0.589	0.443	1.000
Catholic	Other Religion	-0.769	0.515	2.224	0.136	0.680
Catholic	Protestant	-0.006	0.200	0.001	0.975	1.000
No Religion	Other Religion	-0.939	0.553	2.883	0.090	0.537
No Religion	Protestant	-0.176	0.283	0.388	0.533	1.000
Other Religion	Protestant	0.762	0.544	1.961	0.161	0.680

Note: Model 1 (offset only), same analytic sample as Table 1, ego-clustered vcov. Catholic-vs-X rows are Catholic minus X. Holm-adjusted across all six contrasts.

Panel D: Wald tests of pairwise equality, pooled log-odds coefficients (Model 3)

Group 1	Group 2	Log-Odds Diff. (G1 − G2)	SE	χ^2	p	Holm p
Catholic	No Religion	0.166	0.215	0.597	0.440	1.000
Catholic	Other Religion	-0.757	0.510	2.205	0.138	0.688
Catholic	Protestant	-0.007	0.202	0.001	0.973	1.000
No Religion	Other Religion	-0.923	0.540	2.918	0.088	0.526
No Religion	Protestant	-0.173	0.279	0.384	0.535	1.000
Other Religion	Protestant	0.750	0.543	1.909	0.167	0.688

Note: Identical contrasts to Panel C, on the Model 3 specification (structural foci, demographic homophily, salience, and ego/alter degree controls). Holm-adjusted across all six contrasts.

Table 5: Sensitivity Analysis: Ego-Level Cluster Bootstrap and Wald Tests, Nominated-Alter Pool Opportunity Baseline (Waves 3–8, 2,000 replicates).

Panel A: Group-level excess in-group tie rate (descriptive)

Group	*n* Egos	*n* Ties	Observed Rate	Baseline Rate	Excess (95% CI)
Catholic	348	7,630	0.835	0.797	0.038 (0.021, 0.055)
No Religion	52	850	0.226	0.089	0.138 (0.083, 0.199)
Other Religion	22	381	0.155	0.026	0.125 (0.027, 0.275)
Protestant	50	1,185	0.181	0.088	0.094 (0.049, 0.146)

Note: Baseline: nominated-alter pool (sensitivity). The excess is a descriptive quantity: for a fixed in-group odds ratio it varies with the group's baseline share, so it should not be compared across groups of different sizes. Panels C–D (log-odds) are the basis for group contrasts. The bootstrap uses the baseline sample (no covariate filtering).

Panel B: Pairwise differences in excess rate (descriptive)

Group 1	Group 2	Mean Diff.	95% CI	Excludes 0
Catholic	No Religion	-0.100	(-0.163, -0.042)	Yes
Catholic	Other Religion	-0.087	(-0.239, 0.013)	No
Catholic	Protestant	-0.056	(-0.109, -0.007)	Yes
No Religion	Other Religion	0.013	(-0.150, 0.138)	No
No Religion	Protestant	0.044	(-0.030, 0.126)	No
Other Religion	Protestant	0.031	(-0.086, 0.192)	No

Note: Unique egos are resampled with replacement, separately within each group, so that each resampled ego's ties move together. Panel B differences are valid because each group's replicates are drawn independently.

Panel C: Wald tests of pairwise equality, pooled log-odds coefficients (Model 1)

Group 1	Group 2	Log-Odds Diff. (G1 − G2)	SE	χ^2	p	Holm p
Catholic	No Religion	-0.553	0.221	6.286	0.012	0.061
Catholic	Other Religion	-1.694	0.518	10.713	0.001	0.006
Catholic	Protestant	-0.479	0.199	5.806	0.016	0.064
No Religion	Other Religion	-1.141	0.554	4.237	0.040	0.079
No Religion	Protestant	0.074	0.281	0.069	0.793	0.793
Other Religion	Protestant	1.215	0.546	4.948	0.026	0.078

Note: Model 1 (offset only), same analytic sample as Table 1, ego-clustered vcov. Catholic-vs-X rows are Catholic minus X. Holm-adjusted across all six contrasts.

Panel D: Wald tests of pairwise equality, pooled log-odds coefficients (Model 3)

Group 1	Group 2	Log-Odds Diff. (G1 − G2)	SE	χ^2	p	Holm p
Catholic	No Religion	-0.553	0.214	6.654	0.010	0.049
Catholic	Other Religion	-1.684	0.509	10.927	<0.001	0.006
Catholic	Protestant	-0.479	0.201	5.682	0.017	0.069
No Religion	Other Religion	-1.131	0.540	4.390	0.036	0.079
No Religion	Protestant	0.074	0.278	0.070	0.791	0.791
Other Religion	Protestant	1.204	0.542	4.933	0.026	0.079

Note: Identical contrasts to Panel C, on the Model 3 specification (structural foci, demographic homophily, salience, and ego/alter degree controls). Holm-adjusted across all six contrasts.

Table 6: Sensitivity to the Opportunity Baseline (Waves 3–8, ego-clustered SEs).

Panel A: Single non-Catholic indicator (headline test)

Baseline	Model 1	Model 2	Model 3
Cohort (primary)	0.035 (0.165)	0.032 (0.164)	0.034 (0.164)
Alter pool	0.633*** (0.169)	0.630*** (0.167)	0.632*** (0.167)
Leave-own-ego-out pool	0.638*** (0.170)	0.634*** (0.169)	0.636*** (0.169)

Panel B: Group log-odds contrasts with Catholic under each baseline

Group	Baseline	Model 1	Model 3
No Religion	Cohort (primary)	-0.170 (0.222)	-0.166 (0.215)
	Alter pool	0.553* (0.221)	0.553** (0.214)
	Leave-own-ego-out pool	0.556* (0.222)	0.556* (0.216)
Other Religion	Cohort (primary)	0.769 (0.515)	0.757 (0.510)
	Alter pool	1.694** (0.518)	1.684*** (0.509)
	Leave-own-ego-out pool	1.724** (0.532)	1.716** (0.524)
Protestant	Cohort (primary)	0.006 (0.200)	0.007 (0.202)
	Alter pool	0.479* (0.199)	0.479* (0.201)
	Leave-own-ego-out pool	0.482* (0.200)	0.482* (0.203)

Panel C: Break-even baseline share (Model 1, pool coefficients)

Group	Pool share	Cohort share	Model 1 (pool)	Break-even share
No Religion	9.0%	12.4%	0.553	10.5%
Other Religion	2.7%	4.6%	1.694	9.3%
Protestant	8.9%	9.9%	0.479	9.9%

Panel D: Free offset coefficient (pool baseline; Model 1 covariates)

Estimate	SE	p vs. 0	p vs. 1
0.883	0.275	0.001	0.671

Note: Cohort = Wave 1 religious composition of all egos with religion known (time-invariant). Pool = wave-specific composition of nominated student alters, which is partly the outcome. LOO pool = pool excluding each ego's own nominations. Panel A reports the coefficient on a single non-Catholic indicator. Panel C break-even shares hold the Catholic share at its cohort value and use group-mean offsets, so they are approximate. Panel D is identified only under the pool baseline. $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

Table 7: Inference Robustness for the Pooled Group Coefficients under Both Opportunity Baselines (Waves 3–8).

Panel A: Model 1 group coefficients (log-odds vs. Catholic)

Baseline	Group	Est.	SE (ego)	SE (two-way)	Boot 95% CI	p (ego)	p (two-way)
Cohort	No Religion	-0.170	0.222	0.234	(-0.645, 0.240)	0.443	0.466
	Other Religion	0.769	0.515	0.518	(-0.439, 1.685)	0.136	0.138
	Protestant	0.006	0.200	0.221	(-0.413, 0.387)	0.975	0.978
Pool	No Religion	0.553	0.221	0.232	(0.088, 0.979)	0.012	0.017
	Other Religion	1.694	0.518	0.520	(0.375, 2.557)	0.001	0.001
	Protestant	0.479	0.199	0.220	(0.095, 0.847)	0.016	0.030

Panel B: Model 3 group coefficients (log-odds vs. Catholic)

Baseline	Group	Est.	SE (ego)	SE (two-way)	Boot 95% CI	p (ego)	p (two-way)
Cohort	No Religion	-0.166	0.215	0.227	(-0.669, 0.232)	0.440	0.465
	Other Religion	0.757	0.510	0.512	(-0.527, 1.596)	0.138	0.139
	Protestant	0.007	0.202	0.224	(-0.422, 0.367)	0.973	0.976
Pool	No Religion	0.553	0.214	0.227	(0.120, 0.965)	0.010	0.015
	Other Religion	1.684	0.509	0.512	(0.453, 2.509)	<0.001	<0.001
	Protestant	0.479	0.201	0.223	(0.083, 0.846)	0.017	0.031

Panel C: Pairwise non-Catholic contrasts (log-odds difference)

Baseline	Model	Contrast	Diff.	Boot 95% CI	p (ego)	p (two-way)
Cohort	Model 1	No Religion vs Other Religion	-0.939	(-1.926, 0.424)	0.090	0.090
Cohort	Model 1	No Religion vs Protestant	-0.176	(-0.725, 0.388)	0.533	0.546
Cohort	Model 1	Other Religion vs Protestant	0.762	(-0.537, 1.739)	0.161	0.162
Cohort	Model 3	No Religion vs Other Religion	-0.923	(-1.805, 0.382)	0.088	0.089
Cohort	Model 3	No Religion vs Protestant	-0.173	(-0.751, 0.376)	0.535	0.548
Cohort	Model 3	Other Religion vs Protestant	0.750	(-0.576, 1.663)	0.167	0.168
Pool	Model 1	No Religion vs Other Religion	-1.141	(-2.123, 0.251)	0.040	0.040
Pool	Model 1	No Religion vs Protestant	0.074	(-0.506, 0.617)	0.793	0.799
Pool	Model 1	Other Religion vs Protestant	1.215	(-0.187, 2.168)	0.026	0.026
Pool	Model 3	No Religion vs Other Religion	-1.131	(-2.027, 0.169)	0.036	0.037
Pool	Model 3	No Religion vs Protestant	0.074	(-0.467, 0.613)	0.791	0.798
Pool	Model 3	Other Religion vs Protestant	1.204	(-0.058, 2.083)	0.026	0.027

Panel D: Leave-one-ego-out influence on group coefficients

Baseline and model	Group	Full estimate	Range over ego deletions	Most influential ego
Cohort Model 1	No Religion	-0.170	-0.317 to -0.095	93699 (No Religion, 36 ties)
Cohort Model 1	Other Religion	0.769	0.075 to 0.886	63904 (Other Religion, 35 ties)
Cohort Model 1	Protestant	0.006	-0.104 to 0.059	44927 (Protestant, 34 ties)
Cohort Model 3	No Religion	-0.166	-0.295 to -0.090	93699 (No Religion, 36 ties)
Cohort Model 3	Other Religion	0.757	0.052 to 0.877	63904 (Other Religion, 35 ties)
Cohort Model 3	Protestant	0.007	-0.106 to 0.071	44927 (Protestant, 34 ties)
Pool Model 1	No Religion	0.553	0.410 to 0.629	93699 (No Religion, 36 ties)
Pool Model 1	Other Religion	1.694	0.997 to 1.802	63904 (Other Religion, 35 ties)
Pool Model 1	Protestant	0.479	0.368 to 0.532	44927 (Protestant, 34 ties)
Pool Model 3	No Religion	0.553	0.429 to 0.630	93699 (No Religion, 36 ties)
Pool Model 3	Other Religion	1.684	0.977 to 1.795	63904 (Other Religion, 35 ties)
Pool Model 3	Protestant	0.479	0.366 to 0.543	44927 (Protestant, 34 ties)

Note: Ego-clustered SEs cluster on egoid; two-way SEs cluster on egoid and alterid. Boot 95% CIs are percentile intervals from 1000 ego-cluster pairs replicates (egos resampled with replacement within group). Panel D refits the model 422 times, dropping one ego at a time; the most influential ego is the one whose removal shifts the coefficient most. With 19 Other Religion egos, these checks indicate sensitivity rather than providing exact inference.

Table 8: Missingness, Sample Composition, and Missing-Indicator Sensitivity (Waves 3–8).

Panel A: Missingness by ego group (student, non-family ties, Waves 3–8)

Ego group	Egos	Ties	Alter religion missing (%)	Roommate missing (%)	Same dorm missing (%)	Alter race missing (%)
Catholic	351	8,898	14.3	0.0	0.0	35.5
No Religion	52	1,065	20.2	0.0	0.0	40.4
Other Religion	22	487	21.8	0.0	0.0	34.7
Protestant	50	1,308	9.4	0.0	0.0	35.9

Ego group	Egos with discussion frequency missing (%)	Egos with race outside the four coded categories (%)
Catholic	10.7	5.3
No Religion	14.6	24.7
Other Religion	18.2	9.1
Protestant	11.3	7.0

Panel B: Tie composition by ego group (%)

Ego group	Baseline sample (10,046 ties)	Complete-case analytic sample
Catholic	76.0	78.7
No Religion	8.5	6.7
Other Religion	3.8	3.4
Protestant	11.8	11.2

Panel C: Model 1 coefficients (log-odds vs. Catholic), baseline sample vs. complete-case sample

Baseline	Group	Baseline sample		Complete-case	
		Est. (SE)	p	Est. (SE)	p
Cohort	No Religion	0.104 (0.188)	0.581	-0.170 (0.222)	0.443
	Other Religion	0.716 (0.516)	0.165	0.769 (0.515)	0.136
	Protestant	0.083 (0.181)	0.646	0.006 (0.200)	0.975
Pool	No Religion	0.844 (0.188)	<0.001	0.553 (0.221)	0.012
	Other Religion	1.662 (0.518)	0.001	1.694 (0.518)	0.001
	Protestant	0.573 (0.181)	0.002	0.479 (0.199)	0.016

Panel D: Missing-indicator models (roommate, dorm, and race missingness coded with indicators; race-not-coded egos retained)

Baseline	Group	Model 2	Model 3
Cohort	No Religion	0.059 (0.184)	0.055 (0.183)
	Other Religion	0.690 (0.524)	0.674 (0.528)
	Protestant	0.072 (0.182)	0.073 (0.183)
Pool	No Religion	0.801*** (0.184)	0.795*** (0.183)
	Other Religion	1.640** (0.524)	1.625** (0.527)
	Protestant	0.562** (0.182)	0.563** (0.183)

Note: The baseline sample has 10046 ties and the complete-case analytic sample has 8562. The missing-indicator sample (9961 ties) keeps ties with missing roommate, dorm, or alter-race information and egos whose race is not one of the four coded categories; each missing item enters with an indicator, and the item is set to 0. Ego-clustered SEs. $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.